

Project Design Phase

Proposed Solution Template

Date	18 June 2026
Team ID	
Project Name	SHOPEZ : E-commerce Application
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Traditional e-commerce websites are often slow, visually dated, lack responsive dark mode alternatives, and complicate user checkout pipelines. Additionally, small-to-medium businesses lack affordable, easy-to-deploy digital storefronts that combine user shopping layouts with a built-in admin dashboard to track inventory, users, and overall sales revenue from a single portal.
2.	Idea / Solution description	ShopEZ is a unified full-stack MERN application. It delivers a glassmorphic, responsive user interface utilizing React 19 + Vite connected to a robust Node.js/Express API and MongoDB database. • Customers benefit from multi-filter catalogs, persistent shopping carts synced to their accounts, self-service order cancellations, and detailed invoices (calculating ₹99 shipping thresholds and 18% GST). • Admins get a dedicated dashboard displaying metrics (total users, products, orders, and sales revenue) along with inventory CRUD systems.
3.	Novelty / Uniqueness	• Glassmorphism Design & LocalStorage Themes: Premium visual interfaces with smooth light/dark mode transitions that persist across user sessions. • Decoupled Security Guard: Segmented user and

		admin schemas with custom JWT-based authorization middleware protecting endpoints
4.	Social Impact / Customer Satisfaction	• Customer Satisfaction: Enhanced user experience through fast page loading (powered by Vite), instant cart total recalculations, clear order statuses, and responsive grids optimized for mobile shopping. • Social Impact: Democratizes e-commerce by providing an open-source, highly performant self-hosted template that local merchants can use to move their physical store inventory online without expensive subscription fees.
5.	Business Model (Revenue Model)	• Direct Retail Sales: Direct markup profits on items listed across categories. • Tiered Shipping Fee Structure: Encourages larger cart sizes by offering free shipping on orders above ₹999, while charging a ₹99 convenience fee for smaller purchases. • SaaS Licensing (Future): Renting the system out to independent local vendors as a white-labeled e-commerce hosting solution.
6.	Scalability of the Solution	• Decoupled Architecture: The frontend React bundle can be cached globally on CDNs (like AWS CloudFront or Vercel) for microsecond loading speeds. • Backend Clustering: Node.js/Express servers can be run in Docker containers and scaled horizontally using load balancers. • Flexible Database Scaling: MongoDB scales dynamically using Mongoose connection pooling, indexing, and cloud-sharding clusters.